TSM 099

Homework 1

Number structure, radicals, and evaluation

Topics 1-8 | Estimated time: about 1 hour

Complete all problems and show your work on separate paper. Plan for about 45-50 minutes of work and 10-15 minutes to check, correct, and submit.

Use available resources to check your answers and reasoning. Complete the self-evaluation in Blackboard, upload a clear photo of your one best representative page, and retain all your written work.

Work without a calculator before checking. Give exact, simplified answers.

Arithmetic foundations

1. Use divisibility rules to decide which of 2, 3, 5, 9, and 10 divide 450 and 234. Explain without long division.

2. Write the prime factorization of 180 and use it to reduce 126/180.

3. Evaluate or state undefined. Explain each zero-denominator case.

$\frac{-42}{-6}, \quad \frac{0}{-9}, \quad \frac{-9}{0}, \quad \frac{0}{0}$

4. Find an integer pair for each sum/product: (a) -1 and -20; (b) -11 and 24; (c) 2 and -15.

5. Simplify.

$\frac{-84}{126}, \qquad \frac{-72}{-120}$

Radicals and signs

6. Simplify exactly.

$\sqrt{72}$

7. Simplify exactly.

$\sqrt{\frac{49}{81}}$

8. Simplify exactly.

$\sqrt{200}$

9. Simplify exactly.

$\sqrt{50} + \sqrt{8}$

Real and complex contexts

10. For each expression, state whether it has a real value. Then simplify over the complex numbers using i where needed.

$$\sqrt{-49}, \qquad -\sqrt{49}, \qquad \sqrt{(-7)^2}$$

11. Simplify for every real x, including negative inputs.

$$\sqrt{16x^2}$$

Order, substitution, and equivalence

12. Evaluate.

$$-3^2 + 2(-4+1)^2$$

13. Evaluate.

$$\frac{18 - 2(3+1)}{5-3}$$

14. Evaluate when x = -2 and y = 3.

$$2x^2 - 3xy + y^2$$

15. Evaluate both values when a = 1, b = -3, and c = -2. Simplify the radical.

$$\frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

16. Evaluate both sides at x = -2, 0, and 3. Then decide, with algebraic justification, whether the equation is an identity.

$$(x-2)(x+2) = x^2 - 4$$

17. A student claims √(a + b) = √a + √b for nonnegative a and b. Test a = 9, b = 16; explain the conclusion.